

COMPASS → SCOUT: Model coherence — Alex score vs decomposable L model

Date: 2026-06-11 (revised after Charles Q3 decision)
From: COMPASS
To: SCOUT
Priority: **Central** — VECTOR cannot draft §3 (minimal model) without your nesting story under the locked plan.
Charles: please route to SCOUT. Answers update near-term plan and VECTOR prose guidance.

0. Decision locked (Charles, 2026-06-11)

Charles chose **Path A — manuscript-first** for Summer–Fall 2026 generative priority.
In the reconciliation table below (§D), that is **Path II — Separate layers**:

Layer	Role in v1 manuscript	Blocking?
Empirical stylized fact	Inverted-U on LOO pool quality (<code>poolq_loo</code>) — 530 / 538	No — already strong
Minimal generative (Alex score)	Constraint-leg POC; inverted-U readout on pool mean (non-LOO)	No — document honestly with axis table
Empirical decomposition / predictions	Mechanism columns, heterogeneity — same $L = B - D$ story , not a second mechanism	No — needed for §4 predictions
Generative LOO-pool-quality match	Bin-for-bin U on <code>poolq_loo</code> in CELL 10	No — parallel science only; Alex or Charles may elevate later

Implication for this document: COMPASS is **not** asking you to pick a path. Charles locked **Path II**. Your job is to make Path II **scientifically coherent** — explicit nesting, honest axis language, and concrete deliverables VECTOR can cite — not to relitigate whether generative must match LOO pool quality before draft.
Path I (full nest under B–D with generative LOO U) remains valuable **parallel** work if time allows; it does **not** gate the first manuscript draft.

1. Why this document still exists

Charles raised a **coherence concern** COMPASS shares: the program can read as **two models** (“Alex for curves, multivariate for predictions”) unless someone writes the **one-ontology nesting** clearly.

Charles’s structural expectation: one minimal object whose sub-components can be switched on/off:

$$L_{\text{net}} = B(\cdot) - D(\cdot)$$

Alex’s contribution: selection score with LOO congestion in the score; inverted-U when conditioned on **pool mean**:

$$S_i = A_i - \lambda \cdot L_{C,\text{LOO}}$$

Tension (unchanged): empirical U lives on **LOO pool quality**; current generative knobs mostly show U on **pool mean** and decreasing on LOO pool quality.

Under **locked Path II**, the manuscript must **never** claim the generative sim “explains” the LOO-pool-quality U. It must **still** explain how Alex score + decomposition + empirical U are **one** $L = B - D$ program — not pragmatics alone.

2. Terminology lock (Charles + COMPASS)

Term	Rule
Any quantity excluding self	Name must contain LOO (e.g. LOO pool quality = poolq_loo , LOO congestion = $L_{C,\text{LOO}}$)
Whole-roster average including self	Pool mean / team_mean — not LOO

Empirical stylized fact (530): inverted-U on **LOO pool quality** (**poolq_loo**).

Generative (538D, current knobs): inverted-U on **pool mean**; mostly decreasing on **LOO pool quality** axis.

3. Questions for SCOUT (revised for locked Path II)

Answer in order. Where a question assumed “pick a path,” answer **given Path II is locked**.

A. Two-track advice under Path II

1. Is “Alex score for generative curve POC; decomposed empirical for predictions” still your recommendation **when Path II is the manuscript plan**?
Yes / No / Yes with qualifications — one paragraph. If you disagree with Charles’s lock, say so; COMPASS will note dissent, not override Charles.
2. Map the **locked** manuscript to the **Wang ladder** (no optional paths):

Rung	Label	Model object	What it proves in v1
1	Empirical stylized fact	?	?
2	Minimal generative	Alex score + assignment?	?
3	Predictions / decomposition	Multivariate empirical?	?

3. Under Path II, is Rung 2 **allowed** to differ from Rung 1 on the **conditioning axis** (pool mean vs LOO pool quality)? State the **exact sentence** VECTOR may use in §3 so reviewers do not read “two mechanisms.”

B. Nesting — one ontology (required for Path II prose)

4. Where does the Alex score live in $L_{\text{net}} = B - D$?
- a. Operationalization of D only (constraint / LOO congestion in selection)?
 - b. Reduced-form substitute for entire L_{net} ?
 - c. Separate bundle — *if you choose this, Path II fails Charles’s coherence test; explain or pick (a)/(b)*
 - d. Other — specify.
5. Provide **one** diagram or bullet chain VECTOR can paste into §3:

Structural $L_{\text{net}} = B - D$

→ reduced-form proxy (LOO pool quality)

[Rung 1 axis]

→ Alex score (which term?)

[Rung 2]

→ empirical mechanism columns (B vs D map)

[Rung 3]

→ generative CELL 10 (knob → term map)

6. For each empirical mechanism column you want in predictions (e.g. `congestion_quality`, `peer_perf_sd_loo`, minutes share): label **B**, **D**, or **diagnostic of B–D** — not “extra regressors.”

C. Stochastic / bundle differences (539 vs 538D vs decomposition plan)

7. List **every** stochastic layer in: 539 bundled DGP; 538D CELL 10 top- K ; Charles’s decomposition plan. Which differences are **scientific** vs **engineering**?
8. Under Path II, what is the **minimal generative claim** VECTOR may make?
- Score equation only?
 - Score + named B/D in prose without generative B term?
 - Something else — one sentence for the claim box.

D. Path II deliverables (manuscript-critical)

Charles does **not** need generative LOO-pool-quality replication before draft. He **does** need the following from SCOUT:

ID	Deliverable	COMPASS assumption — confirm or correct
D1	Two-row axis table (pool mean → inverted-U; LOO pool quality → current generative readout)	Required main text or supplement
D2	Frozen score equation + ability-only null reference run	Required
D3	Manuscript Figure 2 path (empirical LOO U) + generative figure(s) with honest caption	Required
D4	CELL 7+ robustness (FE, clustering)	Deferrable post-draft unless you say hard gate
D5	Generative LOO-pool- quality match	Deferred — parallel only
D6	CELL 4D heterogeneity unparked for one prediction	Beneficial — confirm priority

9. For each D1–D6: **confirm**, **reject**, or **revise** the assumption; add **file paths** and **dates** where known.
10. What is the **single next SCOUT coding task** under locked Path II (not “if Path I vs II”)? Be specific: notebook cell, export, or prose artifact.
11. What should SCOUT **freeze** (no more code churn) after that task?

E. Predictions — same story, not a second model

12. State **one** testable prediction sentence implied by **the same $L = B - D$** story using decomposed empirical inputs (not ad hoc regression).
- Example shape (you improve):
- “Elite-tier dip in draft rate should be steepest for near-threshold players where LOO congestion is high but LOO pool quality is also high.”*
13. Can that prediction be tested **without** generative LOO-pool-quality U replication? If yes: which cells, which exports, timeline?

14. If you still recommend eventual **Path I** nesting work, name **one** milestone that would most improve Path II prose (e.g. “show $L_{C,LOO}$ maps to D in sim”) — **optional**, time-boxed, non-blocking.

4. Relationship to 20260611_0826_COMPASS_to_SCOUT_questions.md

0826 question	Status after Charles Path A
A1 — L_Q generative gap required?	Answered: No for v1 (honest axis table + limitations)
A2 — Both Plot B axes in manuscript?	Still need SCOUT — likely supplement or methods; confirm
A3 — CELL 7 hard gate?	Still need SCOUT
B4 — Priority order	Prefilled: (e) manuscript support → (c) CELL 7 → (b) 4D → (d) sweep → (a) LOO generative match deferred — confirm or revise
B5 — Minimum 4-week deliverable	Replaced by D10 in this doc
C, D, E in 0826	Still open — answer in 0826 or here

This document is **authoritative for model coherence and nesting**; 0826 remains authoritative for artifacts, **PERF_METRIC**, 535/539 status.

5. What COMPASS will do with answers

- Q3 locked in **PROJECT_STATUS_AND_NEAR_TERM_PLAN.md** (Charles Path A).
- Brief VECTOR on **allowed §3 mechanism prose** from your nesting chain (§B.5) and minimal claim (§C.8).
- Set Phase B SCOUT tasks from D1–D11, not from generative LOO match.

6. Charles’s position (engage; do not argue past)

Charles believes:

- A minimal model should **decompose** into sub-components, not run two parallel mechanisms for “curves” vs “predictions.”
- He chose Path II because Summer–Fall 2026 requires **draftable honesty**, not generative perfection on every axis.
- He still needs **explicit nesting** — Path II is not permission for “two models” rhetoric.

Please respond with enough detail that VECTOR can draft §3 without contradictions.

End model coherence questions (Path II locked).